

SOURCE OF TRUTH

Centro Architecture & Philosophy

Pilot Edition — Revision 7 (Milestone 6)

This document defines the architectural principles and product philosophy that guide every design and development decision in Centro. It is version-controlled in the repository as `ARCHITECTURE.md` and rendered to PDF after every roadmap milestone.

Status	Living document — source of truth, updated at the end of every milestone
Governs	Every design and implementation decision in this codebase
Revision history	See "Document History" at the end

Chapter 1 — Learning First

Every client begins in Learning Mode.

Centro does not require a perfect setup before work begins. Onboarding exists to provide enough information to start working safely, not to perfectly configure every client.

Excel imports, onboarding choices, and existing records create only an Initial Client Profile. From the very first collection cycle, Centro observes reality, learns from recurring behaviour, and continuously improves the profile.

Centro does not guess. It observes, suggests, confirms, and only then learns.

Core principles:

- Start working immediately.
- Learn continuously.
- Never guess when confidence is low.
- Improve from real behaviour, not assumptions.

Chapter 2 — The First Collection Cycle

Every client enters the first month in Learning Mode.

Centro requests the client's documents, waits until the client confirms that the submission is complete, and then treats the collection as one business event to learn from.

The first cycle is not a validation phase. It is a learning phase designed to understand how the client actually works — specifically, which documents that client actually sends, nothing broader.

Chapter 3 — Observe → Suggest → Confirm → Learn

Every meaningful change to a client's document collection profile follows the same lifecycle:

1. **Observe.**
2. **Suggest.**
3. **Confirm** with the client whenever appropriate, through WhatsApp.
4. **Learn.**

Centro never changes a client's profile because of a single event or assumption. A pattern must repeat before Centro treats it as worth asking about, and even then, only the client's explicit confirmation — never the

pattern alone — changes the profile.

Chapter 4 — Learning Without Becoming a Burden

Learning must never create unnecessary work.

Centro should quietly observe whenever possible. It asks questions only if the answer has long-term value and the benefit outweighs the interruption.

Questions must be short, natural, and asked only once — if a client has already declined a suggestion, Centro does not ask about the same thing again.

The accountant reviews exceptions — not routine learning.

Core principle: Centro becomes smarter while becoming quieter.

Chapter 5 — The Living Client Profile

Client profiles are living profiles.

Knowledge comes from Excel, onboarding, collected documents, recurring patterns, and confirmed client responses.

No single source is absolute truth.

Businesses evolve, so the client's document collection profile evolves with them — see Chapter 8 for the exact, permanent boundary of what "evolves" is allowed to mean.

Chapter 6 — Decision Hierarchy

Every decision follows the same hierarchy, most specific and most certain signal first:

1. **Learned Knowledge** — a specific fact this organization has already confirmed about its own clients or data. Checked first because it is the most reliable signal available: it was explicitly confirmed, not inferred, and it is scoped to exactly this organization.
2. **Business Rules** — deterministic, universal rules that apply the same way across every organization (e.g. recognizing standard terminology, hard confidence floors, required-confirmation gates). Used whenever no more specific learned fact exists for this organization yet.
3. **Artificial Intelligence** — a supporting tool, reached only once the above two layers find nothing with sufficient confidence. AI is never the foundation of a decision in Centro.
4. **Human Review** — whenever confidence remains insufficient after all three layers above, a person decides. Centro never guesses.

Why Learned Knowledge outranks generic Business Rules

A previously confirmed, organization-specific fact always outranks a generic rule. This is a deliberate refinement of "business rules first": a rule that is merely generic is less trustworthy than a fact Centro was explicitly told and is now honoring. Business Rules remain the universal floor beneath Learned Knowledge — every organization gets at least this level of correctness even before it has taught Centro anything of its own — but never above a fact that organization has already confirmed.

Chapter 7 — Invisible Automation

The best automation is the one users barely notice.

Routine work happens automatically.

Learning happens continuously.

Notifications are meaningful.

Every interruption must have value.

Success is measured by how much work Centro completes without demanding attention.

Chapter 8 — The Boundary of Learning

Centro learns exactly one thing: which documents should be requested from each client.

This is refined only through observed recurrence and explicit client confirmation, following the Chapter 3 lifecycle exactly — never inferred, never assumed, and never applied from a single occurrence.

Centro does not learn, infer, or adapt:

- The client's underlying business, workflow, or operations.
- The accounting firm's own office policies — including business hours, working days, reminder cadence, or the day a collection period begins.

These remain office policy: configured explicitly by the accountant, once, and changed only by the accountant. Centro never modifies them automatically, regardless of what it observes.

This boundary is permanent. It applies to every learning mechanism in Centro — present and future — not only document collection. A future milestone that would cause Centro to learn or adapt timing, scheduling, or any office policy is out of scope by definition, not a case-by-case judgment call.

Document History

Revision	Change
Pilot Edition	Original 7 chapters established as the architectural source of truth.
Milestone 1	Migrated to a version-controlled Markdown source in-repo (<code>ARCHITECTURE.md</code>), rendered to this PDF after every milestone going forward. Chapter 6 revised to state the Decision Hierarchy as actually implemented (Learned Knowledge checked before generic Business Rules) — see the Centro Implementation Report's Milestone 1 addendum for the full rationale. Added Chapter 8 (The Boundary of Learning) as a permanent constraint, established during roadmap planning for Milestone 6 (Adaptive Document Collection) in response to an early draft that would have violated it.
Milestone 2	Chapter 1/2's "every client begins in Learning Mode" implemented as a real, visible flag (<code>clients.learningMode</code>), ending the first time a client's first Collection Request reaches <code>completed</code> . No timing or duration data is captured — this milestone exists solely to gate Milestone 6's "document appears to no longer be needed" detection, which must never fire before a client has completed even one full cycle.
Milestone 3	Chapter 6's Decision Hierarchy generalized to document classification, its second real implementation (after Epic 4's business-type classifier). A client's own manually-corrected document-naming history (<code>document_learned_patterns</code> , scoped per client) is now checked before the generic deterministic heuristic, which now sits ahead of a documented AI-fallback stub — full parity with the 4-layer pattern.
Milestone 4	Chapter 4/7's "the accountant reviews exceptions, not routine learning" given a real, unified surface: the dashboard's existing per-feature queue pattern extended with a "suggested classification" view (clients auto-classified in the 70-94% band), sitting alongside the pre-existing document-review queue rather than a separate, undiscoverable screen.
Milestone 5	Chapter 3's "Confirm with the client, through WhatsApp" — the one lifecycle step that didn't exist anywhere yet — built as reusable, domain-agnostic infrastructure (<code>pending_confirmations</code>), with a real (not throwaway) manual trigger: an employee can ask a client to confirm a document becomes a standing part of their collection at the exact moment of manual assignment. A real bug was found and fixed during this milestone's own verification.
Milestone 6	Chapter 8's one permitted kind of learning, made real end to end: <code>client_document_requirements</code> tracks per-client additions and removals, confirmed only through Milestone 5's WhatsApp loop, applied only to that one client's future snapshots — the service template itself is never touched. Additions require a genuine second occurrence (never a single event); removals require two consecutive completed cycles missing the same document. A structural constraint (a cycle can never complete while a requirement is unsatisfied) required a new, independently useful capability — waiving a requirement for one cycle — to make recurring-absence detection possible at all. A second bug (ambiguous reply resolution when two confirmations are open at once) was found and fixed during this milestone's own verification.

This document is the governing reference for Centro's architecture. Where any implementation decision conflicts with it, this document takes precedence unless the conflict has been explicitly raised, discussed, and resolved here.